

4.1 Systems of Linear Equations

A linear system has a unique solution when its matrix of coefficients is invertible. The determinant decides invertibility.

$$\det(A) = \sum_{\sigma} \operatorname{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}$$

For a two by two matrix the formula reduces to the familiar

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}$$

and the inverse divides the adjugate by this determinant.